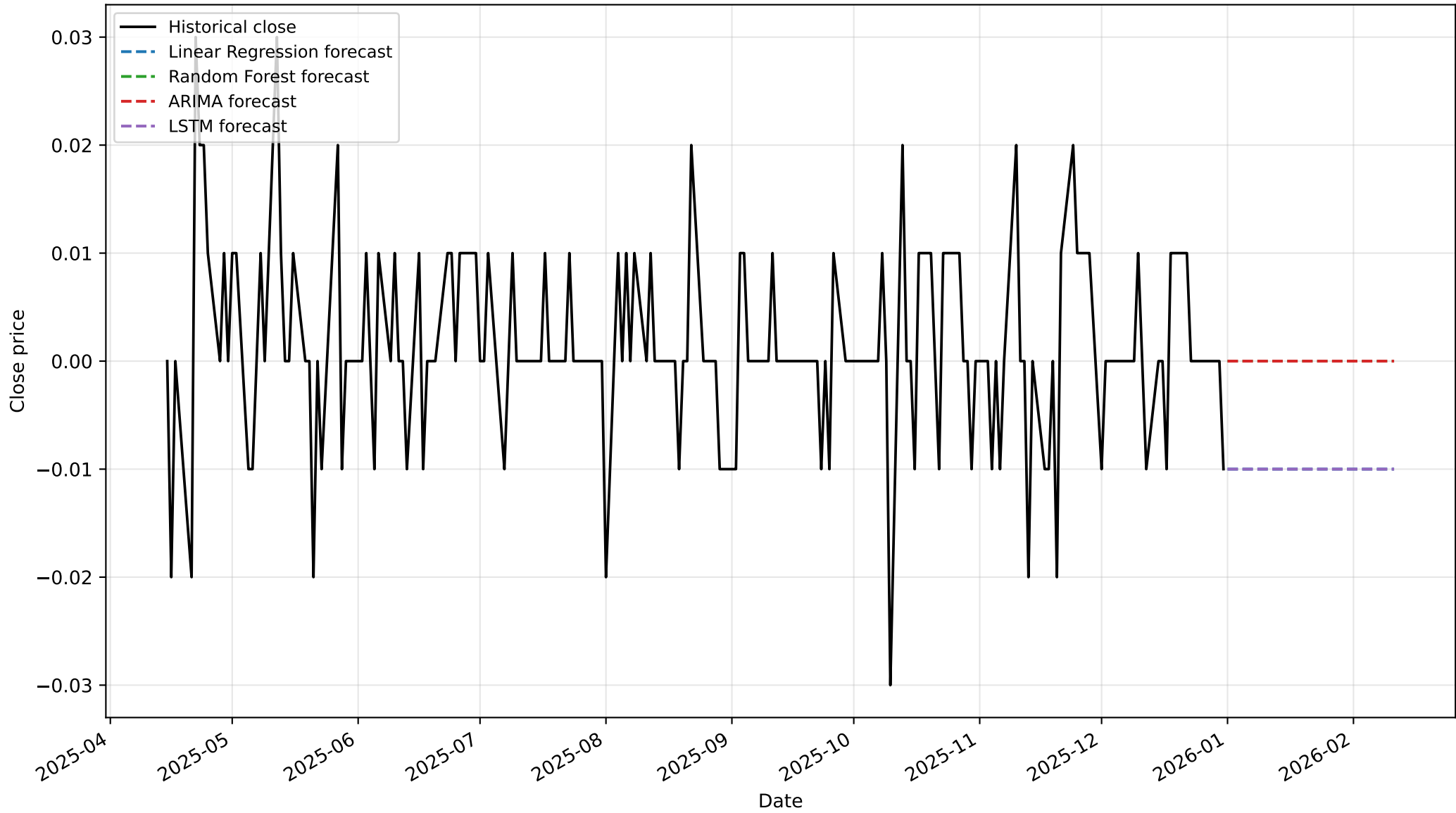


PDA3 Batch Analysis Report
Symbol: ^GSPC
Source file: ^GSPC260102.csv
Generated: 2026-09-28 14:05 UTC
Observations used: 703

Latest indicators
Last Close: -0.01
Change 30D: -19.7
Sma 20: 0.0
Sma 50: 0.0
Volatility: 162390.2

Estimates are educational model outputs, not investment advice. Historical accuracy does not guarantee future performance.

^GSPC — Historical Close and Model Forecasts



Model validation metrics (walk-forward holdout)

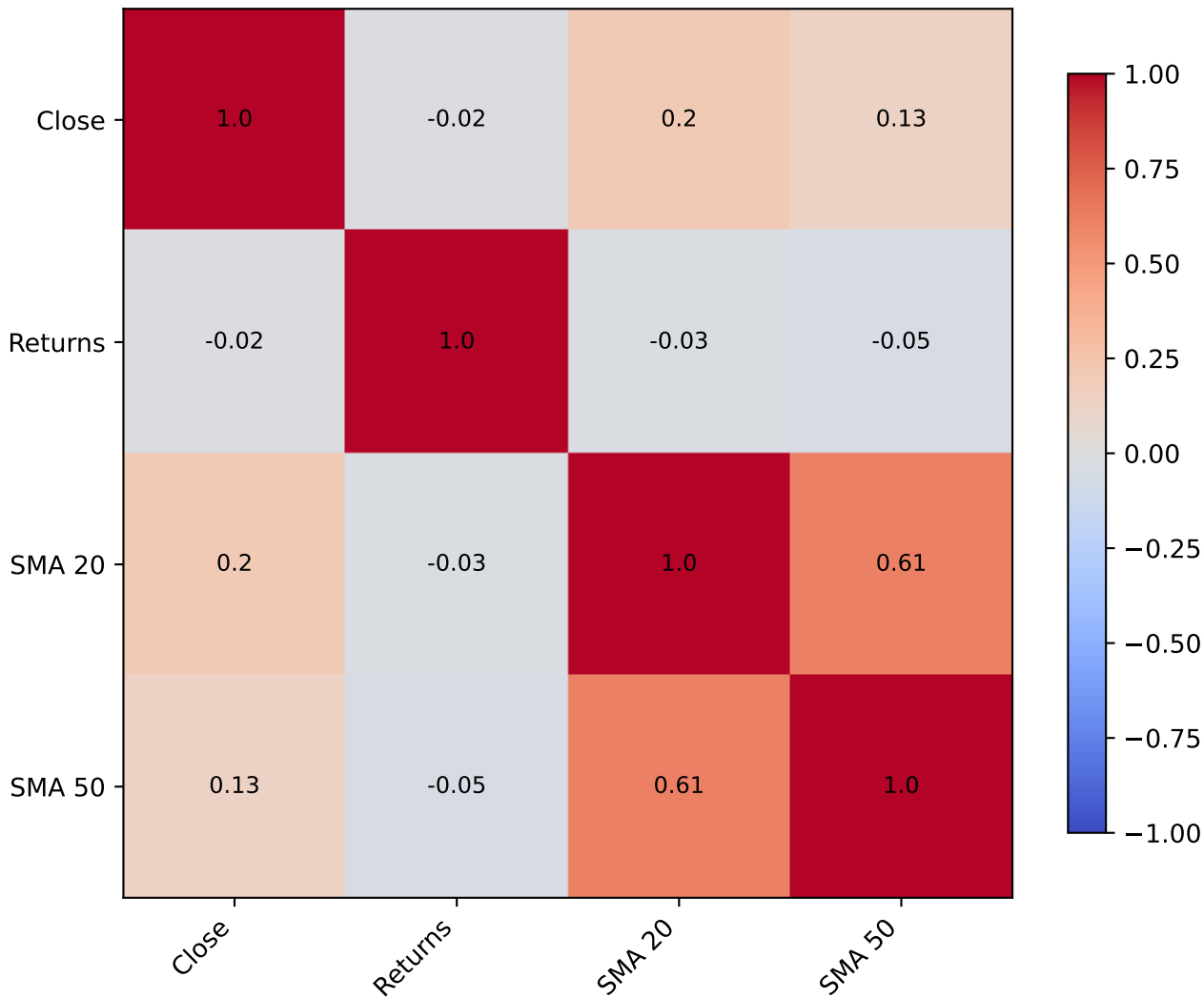
Model	MAE	RMSE	MAPE (%)	R ²	Notes
Linear Regression	0.011	0.012	404.5	-2.2395	
Random Forest	0.005	0.007	111.76	0.0925	
ARIMA	0.006	0.007	99.59	-0.0556	
LSTM	0.01	0.011	334.71	-1.6916	

R² below 0 means a model performed worse than simply predicting the average price over the validation window — common when a stock reverses or plateaus after a strong trend.

Forecast table (next trading days)

Date	Linear Regression	Random Forest	ARIMA	LSTM
2026-01-01	-0.01	-0.0	-0.0	-0.01
2026-01-02	-0.01	0.0	-0.0	-0.01
2026-01-05	-0.01	0.0	0.0	-0.01
2026-01-06	-0.01	0.0	-0.0	-0.01
2026-01-07	-0.01	0.0	0.0	-0.01
2026-01-08	-0.01	0.0	-0.0	-0.01
2026-01-09	-0.01	0.0	0.0	-0.01
2026-01-12	-0.01	-0.0	-0.0	-0.01
2026-01-13	-0.01	-0.0	0.0	-0.01
2026-01-14	-0.01	-0.0	-0.0	-0.01
2026-01-15	-0.01	-0.0	-0.0	-0.01
2026-01-16	-0.01	-0.0	-0.0	-0.01
2026-01-19	-0.01	-0.0	-0.0	-0.01
2026-01-20	-0.01	0.0	-0.0	-0.01
2026-01-21	-0.01	0.0	-0.0	-0.01
2026-01-22	-0.01	0.0	-0.0	-0.01
2026-01-23	-0.01	0.0	-0.0	-0.01
2026-01-26	-0.01	-0.0	-0.0	-0.01
2026-01-27	-0.01	-0.0	-0.0	-0.01
2026-01-28	-0.01	-0.0	-0.0	-0.01
2026-01-29	-0.01	-0.0	-0.0	-0.01
2026-01-30	-0.01	-0.0	-0.0	-0.01
2026-02-02	-0.01	-0.0	-0.0	-0.01
2026-02-03	-0.01	0.0	-0.0	-0.01
2026-02-04	-0.01	0.0	-0.0	-0.01
2026-02-05	-0.01	0.0	-0.0	-0.01
2026-02-06	-0.01	0.0	-0.0	-0.01
2026-02-09	-0.01	0.0	-0.0	-0.01
2026-02-10	-0.01	-0.0	-0.0	-0.01
2026-02-11	-0.01	-0.0	-0.0	-0.01

Correlation matrix

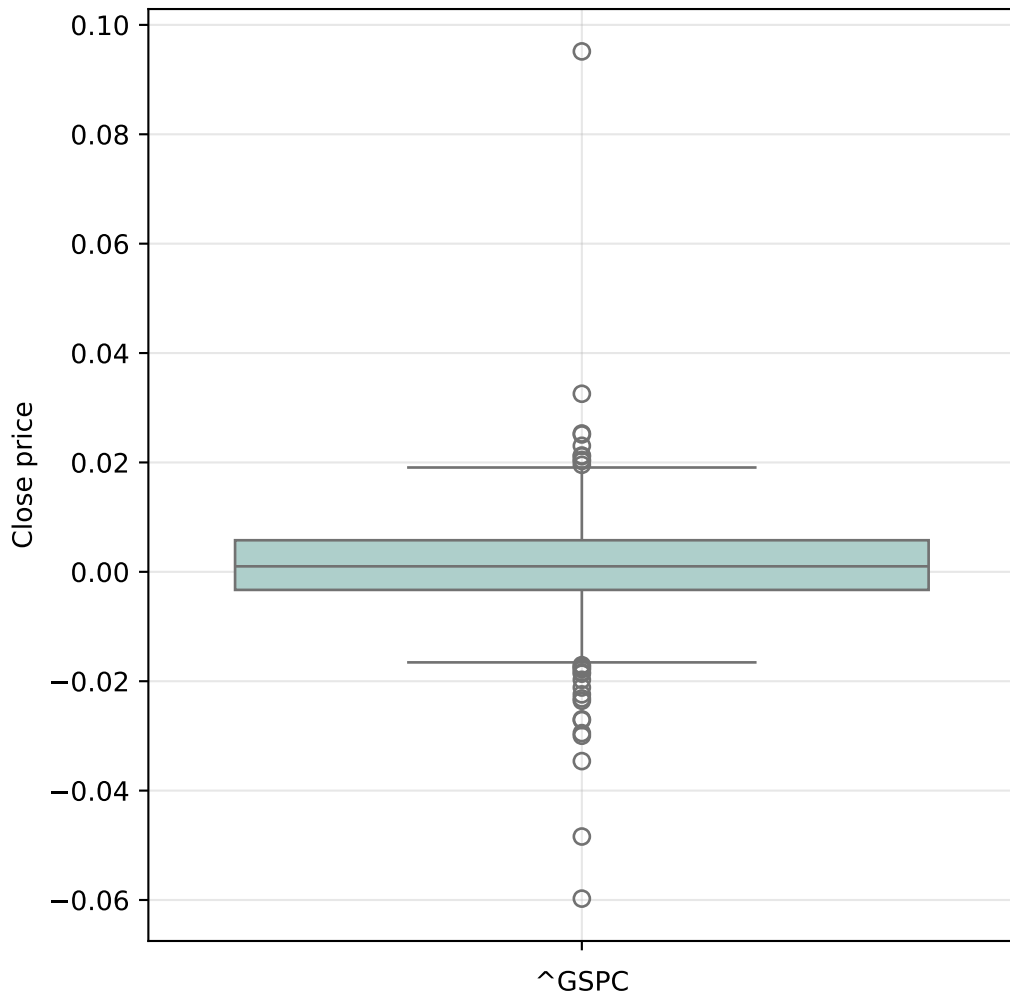


^GSPC — Source data statistics (close price)

Statistic	Value
Count	703
Mean	0.001
Median	0.001
Std Dev	0.009
Min	-0.06
Max	0.095
Range	0.155
Q1 (25%)	-0.003
Q3 (75%)	0.006
IQR	0.009
Skewness	0.62
Kurtosis	17.603
Coeff. of Variation (%)	1123.769

$\hat{\text{GSPC}}$ — Source data range and distribution

Close price boxplot



Close price distribution (violin plot)

